

Some WISE advice

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1 Introduction

In a seminal paper, Moro 2011 argues in favor of an analysis of coordinated *wh*- questions in Italian always requiring *wh*- “movement” operations, also for the coordinated *wh*- item that apparently remains “in-situ” like “perché” (why) in (1):

- (1) Mi chiedo *dove* sono andati *(e) *perché*.
to.me wonder.1SG *where* are gone.3PL and *why/when*
I wonder where they went and why

This configuration gives rise to the so called “*Wh*-in-Situ Effect” (WISE), which is obtained by “*clause structure folding*”, as Moro dubbed the coordination operation targeting a split-CP hosting the two *wh*- phrases. In details, the current linear order is obtained by moving “where” into the (lower part of the split-)CP layer, then base-generating *why* into the higher position of the “split-C” periphery of the clause, obtaining [*perché* C [*dove_i* C [...*t_i*...]...]] as an intermediate step during the derivation of (1). The movement of the lower part of the CP [*dove* C ...] to “spec,&” produces then the final structure (folding) summarized in (2):

- (2) [[*dove_i* C [... *t_i* ...]]_j [[*(e)] [*perché* C *t_j*]]]

2 Two problems

Assuming that this intuition is essentially correct, two problems remain to be fixed under this analysis:

Problem 1, a locality violation of the *wh*- edged clause (*dove* , “where” in (3), which pied-pipes the lower clausal part into the first conjunction position) crossing the higher *wh*- (*perché*, “why”) which remains “in-situ” into the split CP:

(3) [[*dove* C [[...]]]_j [(e) [*perché* C t_j]]]

Problem 2, in contrast with (1), the ungrammaticality of the overt coordination head, in inherently transitive predicates (like “riempire”, fear, or “abbracciare”, hug¹), when only argumental *wh*- phrases are involved (4):

(4) Mi chiedo *chi* abbia riempito (*e)
 to.me wonder.1SG who_{agent/subject} has filled *and
 cosa.
 what_{recipient/object}
 ‘I wonder who filled what.’

3 A tentative proposal

To solve these two issues, here I propose to retain the idea that coordination targets CPs (with interrogative force) and both *wh*-items “move”, but in a reverse derivational perspective. Assuming Phillips’ (1996, 2003) left-right derivation of coordinated structures, I suggest that (i) the “first” CP conjunct is built and processed, incrementally, that is, literally, first, (ii) then the coordination head merges, forcing the “sealing” of the first coordinated phrase, and (iii) when the extra

1. Thanks to Davide Mocci for bringing this contrast to my attention: the opposition is clearly stronger than the milder effect obtained with other predicates that can also appear as unergatives (e.g. “read” or “eat”), as originally proposed.

3.1 *Wh-* items targeting the same left peripheral positions can appear in both conjuncts, *Wh-* items targeting functionally lower positions can only appear in the first conjunct.

Following Rizzi & Bocci 2017, *wh-* modifiers such as “*come*” (*how*) target a ModP position in the split left-periphery, which is higher than the position targeted by the embedded *wh-* item (“*mi chiedo* [*Q_{emb}* *che cosa*] ...”, “*I wonder (that) what* ...”). As predicted, (6).a is possible, (6).b is not (with the conjunction overtly expressed):

- (6) a. Mi chiedo *che cosa* abbiamo letto e
 to.me wonder.1SG *what* have.3PL read and
 come/quando
 how/when
- b. *Mi chiedo *come/quando* abbiamo letto e
 to.me wonder.1SG *how/when* have.3PL read and
 che cosa
 what
 ‘I wonder [*what*] / [*how/when*] they read it and [*how/when*] / [*what*].’

Coherently with this perspective, since the Int(errogative, Rizzi 1997) position related to “why” is higher than any other *wh-* item (Rizzi 1997), the contrast in (1) is readily explained.

On the other hand, free distribution is available with *wh-* modifiers that must be interpreted in the same left-peripheral position (Mod):

- (7) a. Mi chiedo *con chi* lo abbiamo visto *(e)
 to.me wonder.1SG *with whom*, him.CL have.3PL seen and
 dove
 where

- b. Mi chiedo *dove* lo abbiamo visto *(e)
 to.me wonder.1SG *where* him.CL have.3PL seen and
con chi
with whom
 ‘I wonder [with whom] / [where] they saw it and [where] / [with whom].’

The “reconstructed” material (i.e. the complement of C) is fully processed (i.e. interpreted) in the higher position, then it becomes “sealed” and interpretatively unalterable (a part from variable binding necessary to interpret *wh*-modifiers and arguments in their lower position much as in the quantifier binding analysis proposed in Bianchi & Chesi 2010); anything conjoined later can only take as its (elliptical) complement this sealed constituent. This plainly explains:

- (a) the lack of intervention effects (Problem 1): the interrogative force of the first conjunct is interpreted in the first CP; the argumental/functional property of the *wh*-item is interpreted in the clause before it becomes a qualified conjunct able to license elliptical material; then when “e” (*and*) is merged and a new (interrogative) C layer is instantiated, the item that qualifies as its complement simply does not bring any interrogative force to be licensed again (paraphrasing the meaning of the second conjunct in (1), I was wondering *why* “they went *somewhere*”);
- (b) sluicing counter-example discussed in Moro 2011 can now be readily solved excluding (8).a in favor of (8).b:
- (8) a. *Mi chiedo com’è che è partito e quand’è
 to.me wonder.1SG *how*’is that he left and when’is
~~che è partito~~
 that he left
- b. Mi chiedo com’è che è partito e quando
 to.me wonder.1SG *how*’is that he left and when
~~è che è partito~~
 is that he left
 ‘I wonder how it is the case that he left and when.’

Apparent “pseudoclefts” as the one expressed in (8), can be analyzed as full-fledged clefts, along the lines of Pescarini & Donzelli 2018: The focalized *wh*- “how” must fill the functional position related to the modificational interpretation into the (reduced) clausal complement (I assume “he left *in some specific way*”). This seals the copular IP domain and prevents the second conjunct to rely on a parallel cleft construction in which the temporal dimension (*when*) must be integrated yet, in the end, licensing only the full IP complement (“I wonder when [_{IP} it is the case that he left (in this way)]”).

3.2 Extra *wh*- arguments cannot fill a gap into a “sealed” constituent

- (9) a. Mi chiedo *chi* abbia riempito (*e) *cosa*
 to.me wonder.1SG *who* has filled and *what*
 b. *Mi chiedo *cosa* abbia riempito (e) *chi*.
 to.me wonder.1SG *what* has filled and *who*
 ‘I wonder who/*what filled what/*who.’

The contrasts in (9) are crucial for two reasons: on the one hand the order of non D-linked *wh*- complements seems to be rather strict (subj > obj, (9).a, vs. obj > subj (9).b); on the other, the preference for the absence of the conjunction sharply contrasts with the cases in (1), (6) and (7) (Problem 2).

The explanation of these facts relies on the impossibility for the predicative nucleus of the first conjunct to qualify as a legal antecedent for the forthcoming ellipsis (if present) that follows the *wh*-item, if the second conjunct shall occupy a thematic role which must have been already filled before the conjunction is merged. More specifically, the strong preference for the absence of the conjunction (Problem 2) is related to the ambiguous interpretation of this construction and to the preference for the in-situ interpretation of the direct object *wh*- item (no structure folding is assumed in this case): In the missing conjunction case, the proposed analysis predicts that

the first *wh*- item checks the left peripheral *wh*-operator feature, activating the interrogative force of the sentence, while the second *wh*-operator, related to a DP readily integrated into the relevant thematic position, is trivially bound by the higher one. This gives rise to a pair-list interpretation, that is, the expected answers are lists of options of who exactly said what (“A said X” or “B said Y” etc.). On the other hand, when the conjunction is really merged, either a “right-node-raising” (RNR) analysis à la Phillips 2003 is triggered (because of the incompleteness of the thematic shells of the preceding predicate) or a structure folding à la Moro, rephrased in left-right terms as discussed before. This second option, however, would weaken the “impenetrability” condition of the “sealed” (i.e. incrementally fully interpreted) previous constituent (this is because the thematic position must be filled in the first conjunct by a dummy variable, later bound by the object *wh*- in the second conjunct). This formulation of the “impenetrability” condition would, in fact, readily predict the impossibility of (9).b: under the presence of a conjunction, the thematic grid of “fill” will be filled with *pro* (external argument) before *what* be integrated as direct object (internal argument, i.e. the last one in left-right / top-down terms; being the “last argument” is the crucial factor that licenses the RNR analysis). When “and” is merged, the antecedent is then “sealed” and the next *wh*- subject will not be integrated into any thematic position. Notice that the hypothetical analysis based on the binding of the dummy variable in the thematic position would predict this construction to be grammatical, contrary to fact, in the end supporting the RNR analysis.

4 Conclusion

In these pages, I defended Moro’s clause structure folding analysis of multiple *wh*- questions in Italian discussing two related puzzles: the strict ordering of only certain *wh*- items, and the absence of a coordination head with *wh*- arguments. Both puzzles are solved by relying on a reversal, incremental, structure building procedure (top-down, left-right) which seems to me a WISE, compatible, solution.

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